Classical Piano Music Generation Using Deep Learning

Weekly Progress Report – Week 1

Prepared by:

Adham Ahmed 8640

Eyad Ahmed 9073

Hazem Hassan 8619

Belal Mohamed 8906

Project Overview

This project aims to develop a deep learning-based system capable of generating classical piano music using symbolic MIDI data. Instead of working with raw audio, the project uses MIDI representations which encode musical information such as pitch, timing, and velocity. The goal is to train a sequence model (LSTM) that learns musical patterns and generates new piano compositions that resemble classical music.

The system will follow a structured engineering pipeline consisting of data collection, preprocessing, tokenization, model training, music generation, and evaluation. Week 1 focused entirely on dataset preparation and preprocessing.

Key Terms and Definitions

MIDI (Musical Instrument Digital Interface)

A symbolic representation of music that stores notes, pitch, timing, and velocity rather than raw sound.

Pitch

The musical note being played (represented by MIDI numbers 21–108 for piano).

Velocity

Represents how hard a key is pressed (affects loudness).

Quantization

The process of aligning note timings to a fixed rhythmic grid (e.g., 16th notes).

Polyphony

The ability to play multiple notes at the same time (chords).

Chunking

Splitting long musical pieces into fixed-length segments (30 seconds in this project).

LSTM (Long Short-Term Memory)

A type of recurrent neural network designed to model sequential data such as music.

Tokenization

Converting musical events into discrete symbolic tokens for model training.

Week 1 Progress: Data Preparation and Preprocessing

During Week 1, the primary objective was to build a clean and structured dataset for training the music generation model. Two datasets were selected:

- MAESTRO MIDI Dataset (1,276 files)

- Kaggle Classical MIDI Dataset (295 files)

The datasets were inspected for corrupted files, instrument counts, and duration statistics. Both datasets loaded successfully with no corrupted files detected.

Preprocessing Pipeline Implemented

1. Piano Filtering: Kept only piano instruments (MIDI program 0–7).

2. Track Merging: Merged multiple piano tracks into a single note stream.

3. Time Quantization: Snapped notes to a 16th-note rhythmic grid.

4. Pitch Clipping: Limited pitches to the piano range (21–108).

5. Velocity Binning: Reduced velocity to 8 discrete bins.

6. Chunking: Split each piece into 30-second segments.

7. Filtering: Discarded chunks with fewer than 10 notes.

Results of Preprocessing

• MAESTRO → 24,271 processed chunks

• Kaggle → 2,513 processed chunks

• Total → 26,784 clean 30-second piano-only MIDI clips

The preprocessing phase is now complete. The dataset is standardized, clean, and ready for tokenization and model training in Week 2.